Rizal Maulana (Rizal)

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https://rizalmaulanaa.github.io/

EDUCATION

Graduate in Computer Science, Sep 2020 – July 2022
University of Indonesia, Depok, West Java.

Undergraduate in Computer Science, Aug 2016 - Feb 2020
University of Brawijaya, Malang, East Java.

High School, Science Department, June 2013 - June 2016
SMA Laboratorium Percontohan UPI, Bandung, West Java.

WORK EXPERIENCE

Technical Solutions Architect – PT. Indo Koppe Enjiniring (PLN Icon Plus onsite) May 2026 – Present

- Designed AI solution architectures for enterprise use cases involving LLM, OCR, and computer vision technologies.
- Collaborated with stakeholders and engineering teams to translate business requirements into scalable AI solutions.
- Evaluated and proposed AI infrastructure, deployment strategies, and system integration approaches for production environments.

Sr. AI Engineer – PT. Semesta Integrasi Digital Dec 2025 – Apr 2026

- Developed a document comparison system to analyze and identify differences between report versions for accurate version control.
- Designed and evaluated a customer service chatbot using a simple Retrieval-Augmented Generation (RAG) approach, including ticket creation and update functionality.
- Migrated organizational data pipelines using dbt to improve data transformation and modeling workflows.

AI Engineer – PT. Prima Data Semesta (PLN Icon Plus onsite) Mar 2024 – Dec 2025

- Designed and implemented a scalable OCR and Embedding pipeline for nationwide document digitization.
- Developed a household image–based rating and scoring system.
- Designed and deployed a vehicle counting and speed estimation solution using CCTV footage.
- Built and integrated an AI Assistant to support internal PLN document workflows.
- Built robust OCR and LLM-based solutions to extract structured data from Indonesian ID cards.
- Developed and deployed a safety hazard detection system for industrial risk monitoring.
- Performed sentiment analysis on mobile app reviews to identify user concerns and improve experience.
- Engineered data pipelines to support efficient and secure data ingestion.
- Conducted clustering analysis on time series power usage data for consumer behavior segmentation.
- Led the on-premise deployment of large language models for internal enterprise applications.
- Integrated fleet management data to support real-time operational processing.
- Added Speech-to-Speech capabilities to enhance the AI Assistant's interaction experience.

Data Engineer – PT. Sharing Vision (BRI onsite) Mar 2023 – Mar 2024

- Developed and optimized ETL pipelines using SSIS.
- Created interactive reports and dashboards with SSRS.
- Automated data processing workflows for banking analytics.

Data Engineer – Legal Analytics, Telkom Indonesia Sep 2022 – Jan 2023

- Web scraping & data extraction from news portals and social media.
- Built Regex-based document extraction for letters and ID cards.
- Implemented social media crawling for business intelligence insights.

Assitant of Advance Machine Learning – Fasilkom, University of Indonesia Feb 2021 – June 2022

- Instructed students in Python, TensorFlow, and Deep Learning fundamentals.
- Guided students on model training and evaluation technique.

PROJECTS

1. **Speech-to-Speech (STS) System:** Developed a speech-to-speech AI system capable of converting spoken input into translated and re-spoken output.
2. **OHIF Customization:** Customized and extended the open-source OHIF medical imaging viewer to meet the requirements of Indonesian hospital workflows.
3. **Tree Detection:** Built a tree detection model from aerial imagery using a Dilated Residual U-Net architecture.
4. **Indonesian News Clustering:** Applied seed-guided deep document clustering to group Indonesian news articles by topic.
5. **Brain MRI Segmentation:** Designed a system to automatically segment white matter hyperintensities in brain MRI scans using Attention U-Net++.
6. **Newsletter Topic Extraction:** Extracted major topics from newsletters using the CLARA clustering algorithm.
7. **Customer Segmentation Analysis:** Created customer segments for IndiHome users through behavior analysis and K-Means clustering.

Full Project List on GitHub.

TECHNICAL SKILLS

Programming Languages: Python, SQL, SwiftUI
Machine Learning & AI: TensorFlow, Keras, PyTorch
Big Data & ETL: Apache Airflow, Kafka, SSIS, SSRS, RabbitMQ, NATS, Redis
Databases: PostgreSQL, BigQuery, NoSQL (MongoDB, Redis, Elacticsearch)
Cloud & Deployment: Google Cloud Platform (GCP), Docker

PUBLICATIONS & PRESENTATIONS

- **First Author**: Robustness of Probabilistic U-Net for Automated Segmentation of White Matter Hyperintensities (2021)
- **Third Author**: Probabilistic Deep Learning with Adversarial Training and Volume Interval Estimation (2021)
- **Speaker**: Biomedical Image Analysis – Faculty of Computer Science, University of Indonesia

ResearchGate Profile.

SOFT SKILLS

- Strong Analytical & Critical Thinking
- Excellent Problem-Solving Abilities
- Team Collaboration & Leadership